

**SRI SATHYA SAI INSTITUTE OF HIGHER LEARNING
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Paper Code: UCSH-503

Paper Title: Software Lab in Java

ONLINE EXAMINATION SYSTEM

QUESTION.JAVA

```
public class Question {
```

```
private String questionText;
```

```
private String[] options;
```

```
private int correctAnswer;
```

```
public Question(String questionText, String[] options, int  
correctAnswer) {
```

```
    this.questionText = questionText;
```

```
    this.options = options;
```

```
    this.correctAnswer = correctAnswer;
```

```
}
```

```
public String getQuestionText() {
```

```
    return questionText;
```

```
}
```

```
public String[] getOptions() {
```

```
    return options;
```

```
}
```

```
public int getCorrectAnswer() {
```

```
    return correctAnswer;
```

```
}
```

```
}
```

OnlineExam.Java

```
import javax.swing.*;
```

```
import java.awt.*;
```

```
import java.awt.event.*;
```

```
import java.util.ArrayList;
```

```
public class OnlineExam extends JFrame implements ActionListener {
```

```
    private ArrayList<Question> questions = new ArrayList<>();
```

```
    private int currentQuestion = 0;
```

```
    private int[] userAnswers;
```

```
    private JLabel questionLabel;
```

```
    private JLabel timerLabel;
```

```
    private JRadioButton option1;
```

```
    private JRadioButton option2;
```

```
    private JRadioButton option3;
```

```
    private JRadioButton option4;
```

```
    private ButtonGroup buttonGroup;
```

```
    private JButton previousButton;
```

```
    private JButton nextButton;
```

```
    private JButton submitButton;
```

```
private Timer timer;

private int timeLeft = 60;


public OnlineExam() {

    setTitle("Online Examination System");
    setSize(700, 450);
    setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    setLocationRelativeTo(null);

    addQuestions();


    userAnswers = new int[questions.size()];

    for (int i = 0; i < userAnswers.length; i++) {
        userAnswers[i] = -1;
    }

    createGUI();

    displayQuestion();
```

```
startTimer();

setVisible(true);
}
```

```
private void addQuestions() {
```

```
    questions.add(new Question(
        "Which language is used for Android development?",
        new String[]{"Java", "HTML", "CSS", "SQL"},
        0));
```

```
    questions.add(new Question(
        "Which keyword is used to create a class in Java?",
        new String[]{"function", "class", "struct", "define"},
        1));
```

```
    questions.add(new Question(
        "Which component is used to display a single line of text?",
        new String[]{"JLabel", "JButton", "JTextArea", "JFrame"},
        0));
```

```
questions.add(new Question(  
    "Which collection is used to store dynamic elements?",  
    new String[]{"Array", "ArrayList", "String", "Integer"},  
    1));
```

```
questions.add(new Question(  
    "Which keyword is used to inherit a class?",  
    new String[]{"implements", "extends", "inherits", "super"},  
    1));
```

```
questions.add(new Question(  
    "Which method is the entry point of a Java program?",  
    new String[]{"start()", "run()", "main()", "execute()"},  
    2));
```

```
questions.add(new Question(  
    "Which Swing component is used for radio buttons?",  
    new String[]{"JCheckBox", "JRadioButton", "JTextField",  
"JLabel"},  
    1));
```

```
questions.add(new Question(  

```

```
"Which keyword is used to create an object?",  
new String[]{"create", "object", "new", "class"},  
2));
```

```
questions.add(new Question(  
    "Which concept hides the internal details of a class?",  
    new String[]{"Inheritance", "Polymorphism", "Encapsulation",  
"Abstraction"},  
    2));
```

```
questions.add(new Question(  
    "Which class is used to display a message box in Swing?",  
    new String[]{"JOptionPane", "JFrame", "JPanel", "JButton"},  
    0));  
}
```

```
private void createGUI() {
```

```
    setLayout(new BorderLayout());
```

```
    JPanel topPanel = new JPanel(new BorderLayout());
```

```
JLabel titleLabel = new JLabel(  
    "ONLINE EXAMINATION SYSTEM",  
    JLabel.CENTER);  
  
titleLabel.setFont(new Font("Arial", Font.BOLD, 22));  
  
timerLabel = new JLabel("Time Left: 60 seconds");  
timerLabel.setFont(new Font("Arial", Font.BOLD, 16));  
  
topPanel.add(titleLabel, BorderLayout.CENTER);  
topPanel.add(timerLabel, BorderLayout.EAST);  
  
add(topPanel, BorderLayout.NORTH);  
  
JPanel centerPanel = new JPanel();  
centerPanel.setLayout(new BoxLayout(centerPanel,  
BoxLayout.Y_AXIS));  
  
questionLabel = new JLabel();  
questionLabel.setFont(new Font("Arial", Font.BOLD, 16));  
  
option1 = new JRadioButton();  
option2 = new JRadioButton();
```

```
option3 = new JRadioButton();
```

```
option4 = new JRadioButton();
```

```
option1.setFont(new Font("Arial", Font.PLAIN, 15));
```

```
option2.setFont(new Font("Arial", Font.PLAIN, 15));
```

```
option3.setFont(new Font("Arial", Font.PLAIN, 15));
```

```
option4.setFont(new Font("Arial", Font.PLAIN, 15));
```

```
buttonGroup = new ButtonGroup();
```

```
buttonGroup.add(option1);
```

```
buttonGroup.add(option2);
```

```
buttonGroup.add(option3);
```

```
buttonGroup.add(option4);
```

```
centerPanel.add(questionLabel);
```

```
centerPanel.add(Box.createVerticalStrut(20));
```

```
centerPanel.add(option1);
```

```
centerPanel.add(option2);
```

```
centerPanel.add(option3);
```

```
centerPanel.add(option4);
```

```
add(centerPanel, BorderLayout.CENTER);
```

```
JPanel bottomPanel = new JPanel();
```

```
previousButton = new JButton("Previous");
```

```
nextButton = new JButton("Next");
```

```
submitButton = new JButton("Submit");
```

```
previousButton.addActionListener(this);
```

```
nextButton.addActionListener(this);
```

```
submitButton.addActionListener(this);
```

```
bottomPanel.add(previousButton);
```

```
bottomPanel.add(nextButton);
```

```
bottomPanel.add(submitButton);
```

```
add(bottomPanel, BorderLayout.SOUTH);
```

```
}
```

```
private void displayQuestion() {
```

```
Question q = questions.get(currentQuestion);
```

```
questionLabel.setText(  
    "Question " + (currentQuestion + 1) +  
    " of " + questions.size() +  
    ": " + q.getQuestionText());
```

```
String[] options = q.getOptions();
```

```
option1.setText(options[0]);
```

```
option2.setText(options[1]);
```

```
option3.setText(options[2]);
```

```
option4.setText(options[3]);
```

```
buttonGroup.clearSelection();
```

```
if (userAnswers[currentQuestion] == 0) {  
    option1.setSelected(true);  
} else if (userAnswers[currentQuestion] == 1) {  
    option2.setSelected(true);  
} else if (userAnswers[currentQuestion] == 2) {  
    option3.setSelected(true);
```

```
} else if (userAnswers[currentQuestion] == 3) {  
    option4.setSelected(true);  
}
```

```
previousButton.setEnabled(currentQuestion > 0);  
nextButton.setEnabled(currentQuestion < questions.size() - 1);  
}
```

```
private void saveAnswer() {  
  
    if (option1.isSelected()) {  
        userAnswers[currentQuestion] = 0;  
    } else if (option2.isSelected()) {  
        userAnswers[currentQuestion] = 1;  
    } else if (option3.isSelected()) {  
        userAnswers[currentQuestion] = 2;  
    } else if (option4.isSelected()) {  
        userAnswers[currentQuestion] = 3;  
    }  
}
```

```
private void startTimer() {
```

```
timer = new Timer(1000, new ActionListener() {
```

```
    @Override
```

```
    public void actionPerformed(ActionEvent e) {
```

```
        timeLeft--;
```

```
        timerLabel.setText("Time Left: " + timeLeft + " seconds");
```

```
        if (timeLeft <= 0) {
```

```
            timer.stop();
```

```
            JOptionPane.showMessageDialog(
```

```
                OnlineExam.this,
```

```
                "Time is over! Exam will be submitted automatically.",
```

```
                "Time Over",
```

```
                JOptionPane.WARNING_MESSAGE);
```

```
            submitExam();
```

```
        }
```

```
    }
```

```
});
```

```
timer.start();  
}
```

```
private void submitExam() {
```

```
    saveAnswer();
```

```
    if (timer != null) {  
        timer.stop();  
    }
```

```
    int correctAnswers = 0;
```

```
    for (int i = 0; i < questions.size(); i++) {
```

```
        if (userAnswers[i] == questions.get(i).getCorrectAnswer()) {  
            correctAnswers++;  
        }  
    }
```

```
int totalQuestions = questions.size();  
int wrongAnswers = totalQuestions - correctAnswers;
```

```
double percentage =  
    ((double) correctAnswers / totalQuestions) * 100;
```

```
String result;
```

```
if (percentage >= 40) {  
    result = "PASS";  
} else {  
    result = "FAIL";  
}
```

```
String message =  
    "EXAM RESULT\n\n" +  
    "Total Questions : " + totalQuestions + "\n" +  
    "Correct Answers : " + correctAnswers + "\n" +  
    "Wrong Answers  : " + wrongAnswers + "\n" +  
    "Percentage    : " + String.format("%.2f", percentage) + "%\n" +  
    "Result       : " + result;
```

```
JOptionPane.showMessageDialog(
    this,
    message,
    "Final Result",
    JOptionPane.INFORMATION_MESSAGE);

System.exit(0);
}

@Override
public void actionPerformed(ActionEvent e) {

    if (e.getSource() == previousButton) {

        saveAnswer();

        if (currentQuestion > 0) {
            currentQuestion--;
            displayQuestion();
        }

    } else if (e.getSource() == nextButton) {

        saveAnswer();
```

```
        if (currentQuestion < questions.size() - 1) {  
            currentQuestion++;  
            displayQuestion();  
        }  
  
    } else if (e.getSource() == submitButton) {  
  
        int choice = JOptionPane.showConfirmDialog(  
            this,  
            "Are you sure you want to submit the exam?",  
            "Confirm Submission",  
            JOptionPane.YES_NO_OPTION);  
  
        if (choice == JOptionPane.YES_OPTION) {  
            submitExam();  
        }  
    }  
}
```

```
public static void main(String[] args) {
```

```
SwingUtilities.invokeLater(() -> {  
    new OnlineExam();  
});  
}  
}
```

OUTPUT:

```
=====
                        ONLINE EXAMINATION SYSTEM
                        Time Left: 60 seconds
=====

Question 1 of 10:
Which language is used for Android development?

o Java
o HTML
o CSS
o SQL

[ Previous ] [ Next ] [ Submit ]

=====
```

=====

EXAM RESULT

=====

Total Questions : 10

Correct Answers : 8

Wrong Answers : 2

Percentage : 80.00%

Result : PASS

[OK]

=====

=====

TIME OVER

=====

Time is over!

Exam will be submitted automatically.

[OK]

=====